

Project ATHENA — Day 3

Theme: Risk First, Returns Second

Objective: Learn why the best quantitative traders obsess over **risk management**, not prediction accuracy.

Time Required: ~40 minutes

The Biggest Surprise in Trading

Most beginners think:

"If I can predict the market correctly 70% of the time, I'll make money."

Not necessarily.

Now consider this trader:

- Wins **40%** of trades.
- Loses **60%** of trades.

Would you expect them to be profitable?

They absolutely can be.

The key is understanding **expectancy**, not just win rate.

The Concept of Expectancy

Every trading strategy has an average expected outcome.

A simple formula is:

Expectancy =

(Win Rate × Average Win)

–

(Loss Rate × Average Loss)

Example A:

- Win rate = 80%
- Average win = ₹100
- Average loss = ₹600

$0.8 \times 100 = 80$

$$0.2 \times 600 = 120$$

Expectancy = -40

You win often—but lose money overall.

Example B:

- Win rate = 40%
- Average win = ₹500
- Average loss = ₹200

$$0.4 \times 500 = 200$$

$$0.6 \times 200 = 120$$

Expectancy = +80

You lose more trades than you win, yet the strategy is profitable over many trades.

Lesson: Your goal is **positive expectancy**, not a high win rate.

Why Capital Preservation Comes First

Imagine two traders.

Trader A

Starts with ₹100,000.

Loses 50%.

Remaining capital:

₹50,000

To recover, they need:

100% return

Trader B

Starts with ₹100,000.

Loses 10%.

Remaining:

₹90,000

Needed to recover:

11.1%

Small losses are easy to recover from.

Large losses are mathematically expensive.

The First Rule of ATHENA

Before asking:

"How much can I make?"

Always ask:

"How much can I lose?"

Every future strategy in Project ATHENA must define:

- Entry conditions
- Exit conditions
- Maximum acceptable loss
- Position size
- Portfolio impact

If any of these are undefined, the strategy isn't ready.

Risk Per Trade

Professional traders often risk a small fraction of their capital on any single position.

Example:

Portfolio value:

₹10,00,000

Risk per trade:

1%

Maximum loss allowed:

₹10,000

Notice the focus isn't on how much to invest—it's on how much can be lost if the trade fails.

This distinction is fundamental.

How ATHENA Will Use Risk

Eventually, your platform won't simply recommend a stock.

It will answer questions like:

Expected Return: +12%

Maximum Historical Drawdown: -8%

Volatility: Moderate

Suggested Position Size: 3% of portfolio

Risk Score: 6.5 / 10

This is much more useful than a simple "Buy" signal.

Practical Exercise (~40 minutes)

Part 1 — Build Your First Risk Calculator (25 minutes)

Create a new file:

`src/risk/risk_calculator.py`

Write a function that calculates:

- Portfolio value
- Risk percentage
- Maximum permissible loss per trade

Then extend it to calculate position size given:

- Entry price
- Stop-loss price
- Risk amount

Example:

Portfolio = ₹500,000

Risk = 1%

Entry = ₹1,000

Stop Loss = ₹950

Risk per share = ₹50

Maximum portfolio risk = ₹5,000

Position size = 100 shares

This is one of the first genuinely useful building blocks of ATHENA.

Part 2 — Apply It to Your Watchlist (15 minutes)

Using the watchlist you created yesterday:

For each stock, estimate:

- Entry price (current market price)
- A hypothetical stop-loss
- Position size using your calculator

Don't worry about choosing the "perfect" stop-loss. The goal is to understand the mechanics.

Build Log

Update your README.md with a new section:

Components Completed

✓ Project structure

✓ Research journal

✓ Watchlist

✓ Risk calculator

This running log will become a record of your platform's evolution.

Review Question

You have:

- Portfolio = ₹1,000,000
- Risk per trade = 1%
- Entry = ₹2,000
- Stop-loss = ₹1,950

Question:

1. What is the maximum amount you're willing to lose?
2. What is the risk per share?
3. How many shares can you buy without exceeding your risk limit?

Answers:

1. ₹10,000
2. ₹50
3. 200 shares